

ALLIANCE, NE, USA

WMO: 725635

Lat: 42.057N

Lon: 102.802W

Elev: 1198

StdP: 87.74

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 24044

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.3	-18.7	-25.5	0.4	-21.9	-22.0	0.6	-18.3	15.2	-0.8	14.0	-0.8	2.9	280	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	36.1	18.4	34.0	18.3	32.4	18.1	21.1	29.7	20.3	28.9	19.6	28.2	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	15.8	23.5	17.8	14.8	22.9	17.2	14.2	22.5	67.0	29.5	63.7	28.7	60.9	27.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.8	10.5	DB	-29.1	38.8	3.3	1.8	-31.5	40.1	-33.5	41.1	-35.4	42.1	-37.8	43.4	(4)
(5)			WB	-29.4	22.8	3.3	1.2	-31.8	23.6	-33.7	24.3	-35.5	24.9	-37.9	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-3.8	-2.7	2.7	7.0	12.3	18.9	23.0	21.5	16.2	8.0	1.4	-4.1	(6)	
	DBStd	10.86	6.34	6.37	5.92	5.16	4.80	4.07	3.06	3.29	4.72	5.39	5.85	6.49	(7)	
	HDD10.0	1968	428	356	232	116	33	1	0	0	7	99	259	438	(8)	
	HDD18.3	3969	687	589	484	341	193	42	4	11	94	321	507	697	(9)	
	CDD10.0	1396	0	0	6	25	104	267	404	357	193	37	2	0	(10)	
	CDD18.3	355	0	0	0	0	6	58	150	110	31	1	0	0	(11)	
	CDH23.3	5439	0	0	9	39	202	909	2027	1514	651	83	3	0	(12)	
	CDH26.7	2582	0	0	0	4	60	397	1077	745	278	20	0	0	(13)	
(14) Wind	WSAvg	4.8	4.8	5.0	5.3	5.7	5.2	4.8	4.3	4.3	4.4	4.8	4.6	4.7	(14)	
(15) Precipitation	PrecAvg	393	6	7	15	45	70	73	58	40	33	24	9	6	(15)	
	PrecMax	552	17	18	40	86	238	167	119	74	77	72	22	22	(16)	
	PrecMin	184	1	0	0	5	23	13	17	8	6	0	0	0	(17)	
	PrecStd	80	5	6	12	22	49	40	24	19	20	16	6	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	17.6	24.8	27.1	31.3	37.1	38.8	37.3	35.1	29.1	23.0	16.4	(19)	
		MCWB	6.3	7.1	10.3	12.4	15.8	18.0	18.8	18.3	17.5	13.7	9.9	6.4	(20)	
	2%	DB	12.4	13.8	21.0	24.1	28.0	33.7	36.7	35.0	32.5	25.9	19.1	12.4	(21)	
		MCWB	4.6	5.5	9.1	11.5	15.4	18.0	18.7	17.9	16.4	12.5	8.3	4.9	(22)	
	5%	DB	8.7	10.3	17.6	21.3	25.6	31.3	34.6	32.9	30.1	22.4	16.0	8.7	(23)	
		MCWB	2.9	3.8	7.7	10.6	14.6	17.8	18.8	18.0	15.9	11.3	7.0	2.8	(24)	
	10%	DB	5.9	7.3	14.0	17.7	22.6	28.5	32.6	31.2	27.4	18.9	12.2	5.9	(25)	
		MCWB	1.5	2.1	6.1	9.2	13.6	17.4	18.9	18.1	15.3	10.2	5.3	1.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.9	11.6	14.0	18.5	21.8	22.3	21.7	19.6	15.3	10.6	7.2	(27)	
		MCDB	14.4	16.9	21.3	23.3	26.2	29.0	30.8	30.6	29.1	24.7	21.2	15.6	(28)	
	2%	WB	4.9	5.8	9.7	12.2	16.9	20.3	21.4	20.5	18.4	13.6	8.6	5.1	(29)	
		MCDB	11.7	13.1	19.2	21.6	24.3	27.9	30.3	29.1	27.5	23.0	18.3	11.7	(30)	
	5%	WB	3.1	4.1	8.1	11.0	15.7	19.3	20.6	19.8	17.4	12.1	7.2	3.1	(31)	
		MCDB	8.1	9.4	16.7	19.8	23.1	27.4	29.5	28.3	26.0	20.2	15.4	8.4	(32)	
	10%	WB	1.6	2.4	6.6	9.8	14.3	18.3	20.0	19.0	16.4	10.7	5.4	1.4	(33)	
		MCDB	5.4	6.9	13.8	17.2	21.3	26.5	28.8	27.4	24.8	18.1	11.6	5.4	(34)	
(35) Mean Daily Temperature Range	MDBR		14.7	14.5	16.1	15.6	15.4	16.2	17.0	17.0	17.7	17.0	16.6	15.0	(35)	
	5% DB	MCDBR	18.8	19.2	22.7	22.5	20.6	20.2	20.8	20.2	22.1	23.6	22.8	19.0	(36)	
		MCWBR	11.9	11.7	12.2	11.2	9.4	7.4	6.5	6.6	8.5	11.8	13.1	12.2	(37)	
	5% WB	MCDBR	17.4	17.7	21.3	20.4	17.4	17.2	16.8	16.9	18.3	20.3	21.8	18.2	(38)	
		MCWBR	11.4	11.2	11.9	10.6	8.7	7.9	6.4	6.4	7.7	10.6	12.7	12.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.256	0.280	0.307	0.332	0.341	0.369	0.361	0.322	0.283	0.260	0.250	(40)	
	taud		2.570	2.549	2.512	2.440	2.443	2.443	2.403	2.405	2.495	2.577	2.558	2.544	(41)	
	Ebn at Noon		930	972	977	964	941	929	899	896	915	925	904	895	(42)	
	Edh at Noon		66	79	93	108	111	112	115	111	94	76	66	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.23	3.10	4.30	5.32	6.22	7.29	7.21	6.25	5.05	3.47	2.44	1.87	(44)	
	RadStd		0.12	0.17	0.32	0.31	0.49	0.44	0.25	0.22	0.26	0.31	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page